Task-specific reliability of zero-shot protein scores

Sequence likelihood must be interpreted through endpoint-specific evidence.

A Sequence likelihood score

WT

mut

L = log p(mut | context)
- log p(wt | context)

Context-specific scoring

LLR is a score, not a probability.

B Evidence audit

- Endpoint & native metric
- Score version & row identity
- Control applicability

Single-parent: germline control N/A

C Clinical

4,677 variants

20 genes; same LLR

Macro-AUROC 0.7841

5-fold within-panel calibration

Brier	lower is better
Platt	0.1412
Isotonic	0.1407
Gene prior	0.1517

D Paired DMS

69 paired proteins

3B minus 650M

AUROC difference -0.00015

95% CI [-0.0134, 0.0168]

No gain detected

Not an equivalence test

E Binding / antibody

Archived binding

Spearman, not AUROC

ESM-2 650M WT-marginal

General (non-Ab)	0.222
Ab-antigen	0.079

Row/scorer/resampling unresolved. Same SKEMPI set; archived points only; no model claim.

Native AUROC, Brier and Spearman units; no common reliability scale or external clinical validation.